

Micah Shiels

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Portfolio: <https://micah.es> | Location: Spain | Right to Work: EU & UK (Unlimited)

SUMMARY

Mid Software Engineer with 8 years of total engineering / administration experience, including almost **4 years of focused experience in software development and backend engineering**. I bring a strong foundation in databases, distributed systems, and scalable backend architectures, combined with a deep understanding of software engineering principles. My career began in database administration, which honed my attention to performance, data modeling, and system integrity; skills I now apply daily in building robust software systems. I'm now fully focused on backend/software engineering roles, leveraging my knowledge and engineering discipline to deliver production-quality code and solutions.

EXPERIENCE

Mid Software Engineer | StudyHall AI | July 2022 – February 2026 | Remote (UK-based company)

- Achieved significant performance gains on the VertexAI-based EPUB-processing pipeline by implementing Redis caching, fixing race conditions, optimising DB connections, optimising SQL queries, migrating from Pandas to Polars, and eliminating redundant data loads
- Optimised and maintained independently the core PostgreSQL database, cutting heavy query runtimes by **~90%**
- Key contributor of optimisation and implementation of critical FastAPI-powered data science services (planning, exam & module creation, assessment, reporting) used by thousands of users
- Engineered robust Python pipelines to ingest fundamental educational datasets used by the StudyHall AI app – from questions and answers to resources and school exam metadata. Replacing obsolete VBA scripts, therefore reducing time to ingest content by **~90%**
- Optimised and maintained a TypeScript-based Postgres → MongoDB seeder used by the client team heavily reducing runtime by identifying and refactoring N+1 queries
- Built and maintained internal tools with Retool and Streamlit, streamlining workflows and cutting manual processes
- Assisted and guided junior engineers on best practices, code quality, and system design, raising overall team delivery and reliability

English Teacher (TEFL) | Your Language Club Getafe | October 2021 – July 2022 | Madrid, Spain

- Delivered over **400** English lessons, building deep communication skills and the ability to explain complex concepts clearly – an asset in technical collaboration and documentation

Database Administrator | Ministry of Furniture | October 2017 – October 2021 | Neath, UK

- Maintained and optimised independently two product databases (MySQL & Microsoft SQL Server) with 6,000+ SKUs and 300,000+ variants, reducing annual update time by **~80%** through SQL and VBA automation

SKILLS

- **Computer Languages:** Python, SQL, PL/SQL, TypeScript, JavaScript, VBA, HTML, CSS, Neo4j Cypher
- **Frameworks:** FastAPI, Pydantic, Polars, Pandas, SQLAlchemy, SQLModel, BeautifulSoup, Django, Astro, Tailwind
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Neo4j
- **Tools:** GitHub, Docker, Google Cloud, Sentry, Retool, Streamlit, JIRA
- **Natural Languages:** English (Native), Spanish (C2), Portuguese (A2), German (A1)
- **Certifications:** DELE certified C2 (Mastery) Spanish – Fluent speaker, TEFL certified teacher – Distinction level
- **Interpersonal Skills:** Problem-Solving, Critical Thinking, Teamwork, Communication, Teaching, Known for keeping team spirits and morale high